

Text as Data for Economic Research

The Econometrics of Text

Class 5. Generated Variables, Validation, and Inference
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Plan for Today

- A lens first: **prediction vs estimation** — two legitimate uses with different research-design requirements (Ludwig et al.)
- Where text enters an estimation design: **outcome, regressor, treatment, or control**
- **Threat 1 — discovery**: choosing the measure on the same data you estimate on (Egami et al.)
- **Threat 2 — measurement error**: regressing on $\hat{\theta}$, not θ (Battaglia et al.)
- **The fork**: *validate & debias* (Carlson & Dell) vs. *lean on structure* (Battaglia et al.)
- **Frontiers**: text as outcome and as treatment

Slides + discussion. The mechanics you ran in Problem Set 4.

Where This Sits in the Course

- Classes 1–4 built **measures** from text: dictionaries, classifiers, topics, embeddings, LLM labels.
- Each measure was **validated** against human judgment — our recurring theme.
- **Today:** what happens when that measure becomes an empirical object — for prediction, or as an outcome, regressor, treatment, or control.

The Diagnostic Sequence

We will use the same sequence throughout the class:

1. **Task:** prediction or estimation?
2. **Role:** if estimating, is text the outcome, regressor, treatment, or control?
3. **Threat:** discovery, measurement error, or both?
4. **Observability:** can the target be validated on a representative subset, or is it intrinsically latent?
5. **Fix:** research design, sample splitting, validation-based debiasing, structural correction, or joint estimation?

Prediction or Estimation?

Two Empirical Uses of Model Output

Ludwig, Mullainathan & Rambachan (2025)

Start from a text piece r and linked economic variables (Y_r, W_r) .

Prediction

Use r to forecast Y_r . Evaluate out-of-sample predictive loss.

Estimation

Measure a concept $V_r = f^*(r)$, then estimate how V_r relates to (Y_r, W_r) .

Same model, different target, different validity requirement.

One Corpus, Two Questions

Ludwig et al.'s Congressional-bills example

Each r is the description of a bill.

Prediction question

Can the bill description predict whether the bill passes its chamber?

Estimation question

Measure the bill's policy topic. How is topic related to the sponsor's party or roll-call ideology?

The first targets predictive performance. The second targets a relationship between economic variables.

What Ludwig et al. Distinguish

Within **estimation**, the generated concept $\hat{V}_r$ may appear:

- as the **dependent variable**: $\hat{V}_r = W_r' \beta + u_r$;
- as an **independent variable / regressor**: $Y_r = \alpha \hat{V}_r + u_r$.

They do not separately define a causal text treatment. Treatment is a causal-design role developed by Egami et al. and the later text-treatment literature. A regressor can be predictive, descriptive, or causal depending on the design.

Prediction: Define the Target Population

The model produces $\hat{Y}_r = \hat{m}(r; t)$. We evaluate

$$\hat{L} = \frac{1}{N} \sum_{r \in \mathcal{R}} \ell(Y_r, \hat{m}(r; t)) .$$

Which target population $\mathcal{R}$ should this sample loss represent?

- future bills;
- a new policy domain;
- the historical corpus only.

Expected prediction loss is defined relative to $\mathcal{R}$. State the target population before evaluating the model.

Training Leakage: the Class 4 Warning

Recall **directness** from Class 4:

- a model may appear accurate because it has seen the text or later outcomes;
- this is contamination / look-ahead, not genuine prediction;
- prompt instructions cannot erase information already in the weights.

Ludwig et al. call the research-design requirement **no training leakage**.

Evidence of Leakage

In Ludwig et al.'s applications:

- GPT-4o exactly completes parts of historical bill descriptions;
- it also exactly completes some 2019 financial headlines;
- an LLM reading 2019 earnings calls mentions COVID — future information leaking into a purported real-time exercise.

High historical accuracy may reflect memory rather than generalization.

Prediction Design: Match the Target

The appropriate design depends on the prediction target:

- **Future documents:** use fixed model weights, then evaluate on later documents.
- **Private documents:** establish that the documents were unavailable for model training.
- **A fixed historical corpus:** a random sample can represent that corpus, but does not test performance on unseen documents and may reflect memorization.

For forecasting or generalization, the first two designs are the credible cases.

Estimation: Define the Measurement Target

Let

$$V_r = f^*(r)$$

be the measurement a trained human or domain expert would produce if cost were no constraint.

Example: the major policy topic of a Congressional bill.

The target parameter is defined using V_r , not the model label:

$$\theta^* = \arg \min_{\theta} E [g(V_r, W_r; \theta)] .$$

The Plug-In Step

At scale, replace V_r with the LLM label

$$\hat{V}_r = \hat{m}(r; t),$$

and estimate

$$\hat{\theta}_{\text{plug-in}} = \arg \min_{\theta} \frac{1}{N} \sum_r g(\hat{V}_r, W_r; \theta).$$

Define the measurement error

$$\Delta_r = \hat{V}_r - V_r.$$

Accuracy summarizes Δ_r poorly. Bias depends on how Δ_r co-moves with the variables inside $g(\cdot)$.

Congressional Bills: the Estimation Task

Ludwig et al. ask LLMs to label 10,000 bills by policy area.

Then they estimate relationships such as

$$\text{Health-policy bill}_r = \beta_0 + \beta_1 \text{Democratic sponsor}_r + u_r.$$

Changing the **model** or **prompt** changes the estimated coefficients and t -statistics — sometimes substantially.

The danger is not merely a wrong label. It is a wrong economic conclusion.

Financial Headlines: the Regressor Case

The paper also asks LLMs to label each headline as positive, neutral, or negative.

The generated label enters on the right-hand side:

$$\text{future return}_r = \alpha + \gamma \widehat{\text{sentiment}}_r + q'_r \delta + u_r.$$

Across models and prompts, $\hat{\gamma}$ changes in magnitude, significance, and even direction.

This is Ludwig et al.'s **independent-variable** case — not, by itself, a causal text treatment.

The Question for the Rest of Today

The plug-in step buys scale, but $\Delta_r = \hat{V}_r - V_r$ rides into the estimate. Two questions remain — and they organize the rest of the class:

- **How bad?** How does error in $\hat{V}_r$ bias the estimate, and when does it matter? $\Rightarrow$ the **two threats**.
- **How do we fix it?** What can we validate, and which correction follows? $\Rightarrow$ **restoring valid inference**.

This class is about **estimation**: we pose the problem here and build the answer in the sections that follow.

Text in Empirical Designs

Four Roles for Text in Estimation

A text-derived object can enter an estimating equation in four distinct ways:

1. **Outcome:** did a policy change what people or firms say?
2. **Regressor / exposure:** does measured culture, tone, or uncertainty predict an economic outcome?
3. **Treatment:** what is the causal effect of a feature of the text itself?
4. **Control / confounder:** use text to hold content or context fixed while estimating another coefficient.

The role is defined by the **estimand**, not by the NLP method used to construct the measure.

The Same Pipeline, Different Estimands

Classes 1–4 taught us how to construct $\hat{M}_i = g(T_i)$. Class 5 asks what parameter is estimated after construction:

Role	Schematic equation	Coefficient of interest
Outcome	$\hat{M}_i = \alpha + \beta D_i + u_i$	β
Regressor	$Y_i = \alpha + \gamma \hat{M}_i + u_i$	γ
Treatment	$Y_i(g(T_i))$	causal contrast in $g(T)$
Control	$Y_i = \tau D_i + \gamma \hat{M}_i + u_i$	τ

Error in $\hat{M}_i$ lands differently in each row. That is why role comes before the correction.

Text-Derived Outcome: Previous Classes

Here the treatment or explanatory variable is ordinary data; the response is measured from text.

- **Esposito et al. (Class 1):** exposure to *The Birth of a Nation* shifts patriotic and divisive language in local newspapers.
- **Hansen, McMahon & Prat (Class 2):** estimated FOMC topic shares respond to a transparency reform.
- **ClimateBERT (Class 4):** disclosure categories are outcomes used to study TCFD endorsement and France's mandatory-disclosure regime.

Classical measurement error increases residual variance and reduces **precision**, but does not bias the coefficient. Error correlated with treatment, time, or group biases the estimated policy effect.

Text-Derived Regressor: Previous Classes

Here the text measure is an explanatory variable whose coefficient is itself of interest.

- **Cieslak et al. (Class 3):** FOMC uncertainty measures help explain policy stance, especially when inflation is near or above target.
- **Li et al. (Class 3):** embedding-expanded culture scores are related to operational efficiency, risk-taking, firm value, and other outcomes.
- **Remote work (Class 4):** a classifier-generated label can enter a wage regression — the running BCHS application later today.

Classical measurement error produces **attenuation bias**;
non-classical model error can move the coefficient in either direction.

Text as Treatment

Now the causal variable is a **feature of the text itself**, $Z_i = g(T_i)$ — not merely a score placed on the right-hand side.

- Candidate biographies may vary in whether they mention military service; complaint narratives vary in whether they describe mortgage distress.
- Many texts instantiate the same Z_i , while also changing other wording, facts, and tone. Those are different **versions of treatment**.
- Identification requires either controlled textual variation or assumptions that the other features are independent of Z_i or irrelevant to Y_i .

Egami et al.'s CFPB application illustrates the workflow but is explicit that complaint text is not randomly assigned.

Text-Derived Control / Confounder

Here the coefficient of interest is on another variable, D_i :

$$Y_i = \tau D_i + \gamma M_i + \varepsilon_i, \quad \text{but we observe } \hat{M}_i = g(T_i).$$

- Estimating the effect of *how* a firm communicates, we must hold fixed *what* it communicates about.
- Topic shares, embeddings, or text-based matching can proxy for content and context that would otherwise confound D_i .
- Error in the control leaves **residual confounding**; the resulting bias can land on $\hat{\tau}$, even though γ is not the parameter we care about.

This shares measurement-error machinery with a generated regressor, but the **inferential target and interpretation are different**.

Inside Familiar Empirical Designs

Text measures enter designs you already know:

- **Event study:** FOMC tone as a regressor for yield changes.
- **DiD / panel:** news tone or disclosure content as an outcome.
- **RCT:** open-ended responses as outcomes; message features as treatments.
- **Observational causal design:** topics or embeddings as controls, matching variables, or nuisance functions.

The design solves its usual identification problem only if the generated text object behaves as required. **Measurement and discovery problems ride along.**

One Idea, Two Threats

Once the task and role are clear, every text-derived variable faces two distinct questions — often confused, with different fixes:

1. **Discovery:** did we choose g by looking at the same data used for estimation? $\Rightarrow$ identification and overfitting (Egami et al. 2022).
2. **Measurement error:** even with g frozen, is $\hat{M}_i = g(T_i)$ a noisy proxy for M_i ? $\Rightarrow$ bias whose form depends on the role (Ludwig et al.; Battaglia et al.).

Sample splitting answers the first question. It does *not* answer the second.

Threat 1: Discovering the Measure

The Codebook Function g

Egami, Fong, Grimmer, Roberts & Stewart (2022)

Text is high-dimensional. To use it we compress it with a function

g : document $\longrightarrow$ low-dimensional measure.

g is a **codebook**: “is this biography a lawyer?”, “which of K topics?”. It connects raw text to the concept we care about.

g Is Discovered, Not Known

We almost never know g in advance. We **find it from the data**:

- read a sample and write a codebook (manual);
- supervised learning from hand labels;
- unsupervised topic models / clustering / embeddings.

Iterating between coding rules and documents is intrinsic to the work — and the source of the problem.

The Hazard: One Dataset, Two Jobs

The danger is using the **same documents** to

1. **discover** g , and
2. **estimate** the causal effect.

This couples the measure to the particular sample — and breaks inference in two ways: an *identification* problem and an *estimation* problem.

Immigration Experiment: the Assignment

Egami et al. (2022), experiment 3: 1,094 US MTurk respondents

Respondents were randomly assigned a vignette about the **same offense**: illegal entry into the United States.

Control vignette

The person had never previously been imprisoned.

Treatment vignette

The person had served two prior prison sentences, including one for a violent crime, and had previously been deported.

Random assignment identifies the effect of **criminal-history information** within this experimental sample; MTurk recruitment is not random population sampling.

Immigration Experiment: the Text Outcome

After a closed-ended prison question, every respondent explained what the US government should do and why, in at least two sentences.

Let

$Y_i(d)$ = respondent i 's written explanation under assignment d .

The research question is not only *whether* punishment changes. It is also:

Which considerations does criminal-history information cause people to invoke?

The raw outcome is therefore high-dimensional text, not a pre-specified scalar.

From Responses to a Causal Estimand

A structural topic model supplies a codebook

$$g(Y_i) = (g_1(Y_i), \dots, g_K(Y_i)),$$

where $g_k(Y_i)$ is the share of response i devoted to topic k .

Once g is fixed, the topic-specific estimand is

$$\tau_k = E[g_k\{Y_i(1)\} - g_k\{Y_i(0)\}].$$

Random assignment identifies τ_k by a treated–control difference in means.

But here g is learned from the experimental responses themselves.

The Coded Outcome Breaks SUTVA

Standard **SUTVA**: unit i 's response depends only on *its own* treatment — not on how anyone else was assigned. The raw text $Y_i(D_i)$ obeys this.

But we never analyze Y_i directly. We analyze the **coded** outcome, and g is learned from *everyone's* responses:

$$\pi_i = g_{\mathbf{D}}(Y_i), \quad g_{\mathbf{D}} \text{ fit on } \{Y_j(D_j)\}_{j=1}^N.$$

So π_i depends on the **whole assignment vector $\mathbf{D}$** : unit i 's coded outcome moves when *other* units are re-assigned. **The stability SUTVA buys us holds for Y_i , but not for $g_{\mathbf{D}}(Y_i)$.**

Why Full-Sample Topic Modeling Breaks It

Concretely, learn g from every observed response $Y_i(D_i)$:

- The assignment vector $\mathbf{D}$ determines which potential texts enter the corpus, so the learned codebook is really $g_{\mathbf{D}}$.
- Under an alternative treatment assignment, different potential texts would be observed. Learning g again from that corpus could produce different categories or coding rules.
- We would then compare $\tau_{k,g_{\mathbf{D}}}$ across treatment assignments even though the **outcome definition itself has changed**.

This is about hypothetical reassignments of treatment, not different random seeds used to fit the topic model.

Identification: the FPCILV

Fundamental Problem of Causal Inference with Latent Variables

Because the codebook is g_D , an alternative treatment assignment may yield a different g_D — possibly a different *number* and *meaning* of categories.

- The resulting effect estimates do not target the **same estimand**.
- So bias, variance, and consistency of the estimator are **not even defined** — absent further assumptions or a research design.

Egami et al. name this the **FPCILV**: it arises whenever a latent representation used for inference is itself learned from the data.

Estimation: Fishing

A second problem survives even if we set identification aside. The space of codebooks is huge, and we get to choose.

- Refitting g until something is significant is **fishing** — Egami et al.'s own word.
- It generalizes the familiar trap of *recoding variables to chase significance* — but text is far more flexible, so the search space is vast.
- Worse than usual: without splitting you can redefine *both* the dependent and the independent variable in search of a result.

Overfitting Without Ill Intent

The deeper point: this is **not** only about bad actors.

- Even careful analysts *necessarily* search over g 's to meet **interpretability, interest, fidelity, and tractability**.
- Hand-coding iterates the codebook; supervised models tune the classifier; unsupervised methods adjust parameters. The iteration itself is the leak.
- “I was not p -hacking” is no defense — and **validating g on the same data you discovered it on does not undo the data-adaptivity**.

Only a rule fixed *before* the estimation data can break the dependence — next.

The Fix: Sample Splitting

Break the dependence between discovering g and estimating effects.

- **Training set:** discover, label, and validate g — explore freely.
- **Test set:** lock g , apply it, estimate the effect — *used once*.

In the test set, g no longer depends on the randomization $\Rightarrow$ estimands are well-defined and overfitting cannot survive.

The Workflow

1. **Split** the documents (default 50/50).
2. **Discover** g on the training set.
3. **Validate** g on the training set (close reading, label fidelity).
4. **Apply** g and **estimate** effects on the test set.
5. **Validate** g on the test set — but do *not* revise it.

The cardinal rule: once the test set is opened, g is frozen.

Trade-offs and Portability

- **Efficiency:** splitting spends half the data on each task — discovery power vs. estimation power.
- **Portability:** the *concept* travels, but g is corpus-specific.
Right-wing populism needs a different codebook in the US vs. the UK.
- $\Rightarrow$ compare **signs** across settings, not point estimates: a 0.9 here is not a 0.9 there.

How Egami et al. Implement the Split

In experiment 3, the authors divide the 1,094 responses equally:

1. **Training half**: fit several structural topic models.
2. Inspect and label the topics using training responses only; two researchers label independently, then agree on 11 final topics.
3. **Freeze g** : no relabeling after opening the test responses.
4. **Test half**: infer each response's topic proportions and compare treatment with control for each fixed topic.

What the Immigration Experiment Finds

Back to the immigration experiment — now estimating on the test split

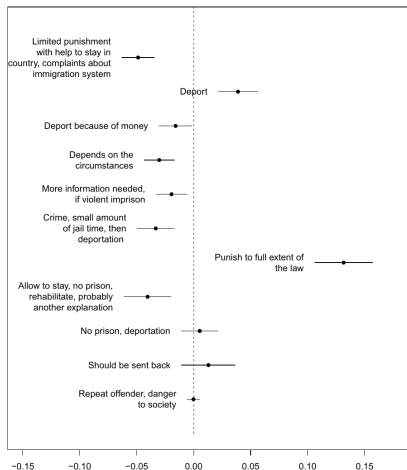
Relative to the no-prior-imprisonment vignette, criminal-history information causes respondents to write more about:

- **maximal punishment;**
- **deportation;**
- **sending the person back** to their country of origin.

Control respondents write more about allowing the person to stay or making punishment depend on the circumstances.

The payoff is causal evidence about the **reasons and frames** behind punishment preferences, not merely a change in a closed-ended answer.

Test-Set Effects by Topic



Left: more prevalent under control.

Right: more prevalent under treatment.

Source: Egami et al. (2022), Fig. 2. Points are treatment-control estimates; lines are 95% confidence intervals.

Threat 2: The Measure Has Error

From Discovery to Error

*Battaglia, Christensen, Hansen & Sacher (2025); applied companion:
Christensen & Hansen (2025)*

Threat 1 was about **choosing** the measure g . Suppose we did it right: split the sample, discovered g on train, froze it.

- The estimand is now well-defined and we are not fishing.
- **But g still applies imperfectly.** The label it assigns, $\hat{\theta}_i$, is a **noisy estimate** of the truth θ_i .

Threat 2: even a perfectly chosen, frozen measure is **measured with error**. What does that do to a regression?

We Regress on $\hat{\theta}$, Not θ

Even with g fixed, the generated variable is a noisy proxy. The target model is

$$Y_i = \gamma^\top \theta_i + \alpha^\top q_i + \varepsilon_i,$$

with θ_i latent (uncertainty, sentiment, a topic share). We observe only $\hat{\theta}_i$.

The naive two-step strategy: generate $\hat{\theta}_i$, then regress Y_i on $\hat{\theta}_i$ and q_i , treating $\hat{\theta}_i$ as ordinary data and ignoring its measurement error.

Outcome vs. Regressor

Ludwig, Mullainathan & Rambachan (2025), Prop. 3

Plugging in $\hat{\theta}_i$ can distort inference — and **where** the measure enters determines how:

$\hat{\theta}$ *as outcome (LHS)*

$\beta = \beta^* + \lambda_{\Delta|W}$ — the projection of the measurement error on the covariates. Mean-independent error gives no bias but lowers precision; systematic error gives $\lambda_{\Delta|W} \neq 0$ and biases β .

$\hat{\theta}$ *as regressor (RHS)*

Attenuation plus error–residual correlation; non-classical error can go *either way*.

Two Sources of Error in $\hat{\gamma}$

Even if the regression model is correct, $\hat{\gamma}$ differs from γ for two reasons:

- **Ordinary sampling error:** with finite n , the estimate varies across samples. Its scale shrinks like $1/\sqrt{n}$.
- **Measurement error:** $\hat{\theta}_i$ may differ from the latent θ_i for each observation.

BCHS ask which source is large **relative to the other**.

Increasingly large datasets paired with increasingly accurate algorithms can leave both first-order.

Running Example: Remote Work and Wages

BCHS application using Hansen et al. (2023) / Lightcast

Hansen et al. use an LLM to classify whether job postings offer remote work. BCHS revisit a downstream wage regression in which this predicted label is a regressor.

For job posting i :

- Y_i is log posted salary;
- $\theta_i \in \{0, 1\}$ is whether the job truly offers remote work;
- $\hat{\theta}_i \in \{0, 1\}$ is the label predicted from the posting text;
- q_i may include occupation fixed effects.

The measurement error is **misclassification**: some postings receive the wrong remote-work label.

The 99% Accuracy Puzzle

The remote-work classifier achieves **98–99% held-out accuracy** and is applied to a corpus containing hundreds of millions of postings.

In BCHS's San Diego accommodation-and-food-services sample, the two-step wage coefficient is nevertheless materially different from the corrected estimate ($n = 16,315$).

The next two slides explain why: accuracy alone uses the wrong denominator. The issue is measurement error relative to the sampling uncertainty of $\hat{\gamma}$.

Error Relative to Sampling

Battaglia, Christensen, Hansen & Sacher (2025)

For an imputed binary label, BCHS study sequences in which

$$\sqrt{n} E \left[\hat{\theta}_i (1 - \theta_i) \right] \rightarrow \kappa \in [0, \infty).$$

$E[\hat{\theta}_i(1 - \theta_i)]$ is the **population share of false positives** (BCHS's expected FPR).

- $\kappa = 0$: measurement error vanishes faster than sampling uncertainty.
- $\kappa > 0$: individually rare errors still shift the first-order distribution.

Classical Error $\Rightarrow$ Attenuation

If $\hat{\theta}_i = \theta_i + u_i$ with u_i classical (mean-zero, independent):

$$\text{plim } \hat{\gamma} = \gamma \cdot \frac{\sigma_{\theta}^2}{\underbrace{\sigma_{\theta}^2 + \sigma_u^2}_{\text{reliability} < 1}}.$$

The estimate is biased **toward zero** — the textbook errors-in-variables case.

Why Attenuation? The Signal-to-Noise View

The reliability ratio is just **signal over signal-plus-noise**:

- When the proxy is mostly signal ($\sigma_u^2 \rightarrow 0$), reliability $\rightarrow 1$ — no bias.
- When it is mostly noise (σ_u^2 large), reliability $\rightarrow 0$ — the coefficient is dragged to zero.

Intuition: noise in a regressor makes it look **less related** to Y than the truth is, so the slope flattens.

This is orientation, not yet the full BCHS result: binary classifier errors are generally **nonclassical**, so the bias depends on the error *structure*, not only its size.

Binary Labels: a Balanced Benchmark

For a stylized case with $P(\theta = 1) = 1/2$, let m be the probability that a binary label is misclassified, assumed equal for true zeros and true ones:

$$\text{plim } \hat{\gamma} = (1 - 2m) \gamma.$$

- The proxy **attenuates** the slope by a factor $1 - 2m$: more misclassification, more shrinkage toward zero.

The balanced-prevalence assumption matters. BCHS do **not** rely on this special formula — they allow asymmetric, nonclassical label error, which is why undoing it needs the next section.

Put Numbers On It

Continue the **balanced benchmark**. Suppose $\gamma = 1.0$ and $m = 10\%$:

- Naive two-step estimate: $\text{plim } \hat{\gamma} = (1 - 0.2) \cdot 1.0 = \mathbf{0.80}$ — a 20% under-statement.
- Worse error, $m = 20\%$: $\hat{\gamma} \rightarrow (1 - 0.4) \cdot 1.0 = \mathbf{0.60}$.
- And a genuinely tiny per-label error still moves the estimate once n is large enough that $\kappa > 0$.

In this stylized case, a “90% accurate” label shaves a fifth off the coefficient — and **nothing in the regression output warns you**.

The Familiar Case First: Pagan (1984)

Familiar case: a common first stage produces a second-stage regressor.

- Examples: expected inflation, a predicted price, a propensity score.
- Shared first-stage estimation **inflates asymptotic variance**.
- Textbook fix: **correct the standard errors**; the estimator remains centered correctly.

In our setting, each $\hat{\theta}_i$ comes from its *own* document. No common first stage $\Rightarrow$ a different problem.

The Surprise: Bias, Not the SEs

The Pagan (1984) intuition says a generated regressor *inflates standard errors*. BCHS show that here it is **misplaced**:

- The asymptotic variance is the **same** as if we had regressed on the true θ_i — so **OLS standard errors are consistent**.
- But the estimator is **centered in the wrong place**.

Two-step CIs have the **right width but the wrong center** — so they under-cover. The primary problem is **bias**, not the SEs.

The Bias, Explicitly

In a stylized no-control binary-label regression, let $p = E[\theta_i]$. BCHS's label specialization implies

$$\sqrt{n}(\hat{\gamma} - \gamma) \rightarrow_d \mathcal{N}\left(-\kappa \gamma \frac{1}{p(1-p)}, V\right).$$

- $\kappa = \lim \sqrt{n} E[\hat{\theta}_i(1 - \theta_i)]$ captures false positives relative to downstream sampling uncertainty.
- V is the *same* variance as the infeasible regression on θ_i .

With controls, the relevant denominator is the residual variation in the true label after partialling out q_i ; the paper uses the general matrix formula.

Reading the Bias Term

The label formula says the centering problem is large when:

- $\kappa = \sqrt{n} \times \text{FPR}$ is **large**: a huge n can make a small false-positive share first-order.
- $p(1 - p)$ is **small**: remote postings are rare, so the true label has little identifying variation.
- $|\gamma|$ is **large**: the location shift scales with the coefficient.

In one (long) line: **even if the false-positive rate is small, a rare true label and a nonzero coefficient can lead to substantial bias; with enormous samples, that bias can also be large relative to sampling uncertainty.**

When Error Is Not Classical

Model error is often **systematic** — correlated with the true θ_i : the classifier misses sarcasm, one domain, one group.

- Then even the **sign** of the bias is unknown: attenuation *or* amplification.
- BCHS show index construction can bias in *either* direction.
- Non-classical error is genuinely harder — the *structure* of the validation errors matters, not only their size — but learn that structure (a full confusion matrix, or a representative sample) and it is **still correctable**.

Class 2 callback: the classifier that keys on a cue correlated with the treatment (tech firms $\rightarrow$ “software”) is exactly this — error that *tracks the treatment*, like an exclusion-restriction violation. No confusion matrix fixes that: it is an *identification* problem, not measurement error.

Index Construction: Bias Either Way

Many text measures classify each document, then **aggregate**: EPU (Baker, Bloom & Davis 2016), geopolitical risk (Caldara & Iacoviello 2022), hawk/dove tone — or the Gentzkow–Shapiro **partisanship gap** you met in Class 2.

- The per-document error need not be classical, so (BCHS §6.3) the bias on the index coefficient can be **positive or negative**.
- The same logic hits a **generated control**: the bias lands on α , not only γ — the confounder case from the start of class.

The Damage — and the Question It Raises

- A generated regressor **biases** $\hat{\gamma}$: the *form* is set by the role, the *size* by κ (error relative to sampling).
- OLS standard errors stay **consistent** — the failure is in the center, not the width, so the regression output never flags it.
- With non-classical error the bias can run **either way**.

Can we undo it? That turns on a single question — can we observe the truth on a validation subset? **That is the next section.**

Restoring Valid Inference

The Fix Depends on the Error You Can Learn

The bias is a **known function of the error structure**. To undo it, you must learn that error — and *how* you learn it sets the method:

1. **Representative validation**: observe the true target on a random subset $\Rightarrow$ debias nonparametrically.
2. **Targeted audit**: observe truth only for predicted positives $\Rightarrow$ an error rate plugged into a structural correction.
3. **Modeled structure**: no external labels $\Rightarrow$ model the error and estimate jointly.

Not a ranking — **pick by what you can see**, from most to least external truth.

Carlson & Dell: The Basic Intuition

Start with the concept we want to measure, such as whether a job truly allows remote work:

$$\text{latent concept} \xrightarrow{\text{valid human rubric}} M_i^* \quad \text{text } U_i \xrightarrow{\text{model}} \hat{M}_i \approx M_i^*.$$

- M_i^* is a high-quality human label that **correctly captures the latent concept**. We observe it only for an annotated subset.
- The model predicts $\hat{M}_i$ for everyone, but may be systematically wrong.
- Human labels provide validity. Full-sample predictions improve precision.

MAR-S corrects $\hat{M}_i$ relative to M_i^* . It cannot repair a bad definition of M_i^* .

Predict Everywhere, Correct on a Subset

Representative validation — Carlson & Dell (2025), MAR-S

Cheap predictions $\hat{M}_i$ on **all** n ; the true M_i^* on a random labeled subset V . For a mean,

$$\hat{\theta} = \underbrace{\frac{1}{n} \sum_{i=1}^n \hat{M}_i}_{\text{prediction on all } n} + \underbrace{\frac{1}{n_V} \sum_{i \in V} (M_i^* - \hat{M}_i)}_{\text{bias, measured on } V}.$$

The first term is cheap but biased; the second **measures that bias** where truth is seen, and subtracts it. **Unbiased** (the terms cancel) and **efficient** (large n makes the first term precise). Extends from this mean to regressions, IV, DiD.

MAR-S: What It Needs and Buys

Missing At Random Structured Data

- **Valid ground truth:** annotation reveals the correctly defined M_i^* using a stable, high-quality rubric.
- **Missing at random:** for annotation indicator A_i ,

$$(U_i, M_i^*) \perp A_i \mid X_i.$$

Conditional on X_i , annotated and unannotated observations are comparable.

- **Buys:** valid inference even with a poor black-box predictor, and greater precision when predictions are informative.

When Full Validation Is Too Costly

Targeted audit — Christensen & Hansen (2025); theory: Battaglia et al. (2025)

A representative validation sample can be expensive — or the target may be intrinsically latent. Then exploit **structure**:

- For a binary label you do not need the whole truth, only the **error rate**.
- Predicted negatives contribute nothing to the bias — a posting the model calls negative cannot be a false positive — so you adjudicate **only the predicted positives**.

Far cheaper than full validation — and the practical core of the Christensen–Hansen guide.

Step 1: Audit the Predicted Positives

The label bias is governed by the **false-positive rate**

$FPR = E[\hat{\theta}_i(1 - \theta_i)]$, the population share predicted remote but truly not.

- Draw a representative random sample; hand-check only the postings the model **predicted remote** (negatives contribute zero).
- Remote-work example: of 1,000 sampled postings, 26 are predicted remote and 9 are false positives $\Rightarrow \widehat{FPR} = 0.009$.

A few hundred trusted human labels on predicted positives — far cheaper than validating all 1,000.

Step 2: Plug the FPR into the Correction

The balanced benchmark of Threat 2 undoes trivially — divide by $(1 - 2m)$. BCHS generalize to **asymmetric** error. For the no-control binary case,

$$\hat{\gamma}_{\text{bca}} = \left(1 + \frac{\widehat{\text{FPR}}}{\widehat{\text{Var}}(\hat{\theta}_i)} \right) \hat{\gamma},$$

with a more aggressive multiplicative version dividing by $1 - \widehat{\text{FPR}}/\widehat{\text{Var}}(\hat{\theta}_i)$.

More false positives, or a rarer positive label, $\Rightarrow$ a **larger** correction. Re-center the CI at $\hat{\gamma}_{\text{bca}}$ (OLS SEs; BCHS also adjust for audit uncertainty). With controls, software uses the general matrix formula.

It Moves the Answer: Remote Work

Effect of remote work on log salary (BCHS 2025, Table 1)

	Two-step	Bias-corrected
Remote-work indicator	0.364 [0.322, 0.406]	0.641 [0.446, 0.836]

The CIs **do not overlap** — even though $\widehat{\text{FPR}} < 1\%$. **Low misclassification can still distort inference materially.**

Model the Error: Joint Estimation

With no audit available, or error too large for a light correction, estimate everything together:

- Treat the text model linking x_i to latent θ_i as an **observation equation** (state-space style).
- Maximize the joint likelihood of the latent-variable model *and* the regression (integrate out θ_i , e.g. via Hamiltonian Monte Carlo).

The most structure, the least external data — it corrects bias *and* improves the measure as a by-product.

Beyond Labels: FOMC Sentiment

Hawk/dove tone $\rightarrow$ *policy path*, 199 meetings (1995–2023)

	Two-step (raw)	Two-step (est.)	Bias-corrected
Hawkish	0.080 [−0.006, 0.165]	0.063 [−0.012, 0.138]	0.090 [0.015, 0.165]
Dovish	−0.048 [−0.159, 0.063]	−0.046 [−0.145, 0.053]	−0.098 [−0.197, 0.001]

The same corrections apply to an **index** (an aggregate score), not just a binary label. Correction shifts the CIs to **exclude zero**; joint estimation roughly triples the effect and R^2 .

Which Tool, and When

- **Representative validation (MAR-S)** — weak assumptions, efficient, any estimand; needs a representative annotated sample.
- **Targeted audit (analytic correction)** — cheapest; needs only an FPR / confusion structure, when the binary or index form is credible.
- **Modeled structure (joint estimation)** — no external labels, strongest assumptions; for latent targets or large error.

More external truth $\Rightarrow$ weaker assumptions. **Pick by what you can see.**

Software Exists

No need to code the corrections by hand — one tool per fix in the sequence:

- **Representative validation** (Carlson & Dell, MAR-S): `mar-s` (R) — `github.com/jscarlson/mar-s`. Debiases a mean, regression, IV, or DiD against a labeled subset.
- **Targeted audit** (BCHS, binary FPR): `ols_bca` (additive) and `ols_bcm` (multiplicative) — the $(1 + \text{FPR}/\widehat{\text{Var}})$ correction.
- **Index / topic-share** via confusion matrix: `ols_bca_topic`.
- **Modeled structure** (no external labels): `one_step` — joint MLE over $\theta_i, \hat{\theta}_i, Y_i$.

BCHS functions ship in `ValidMLInference` (Python) and `MLBC` (R); same syntax as OLS, plus the FPR (or confusion matrix).

The Validation Sample Is the Key Asset

- A **representative** sample of the truth powers MAR-S; even a **targeted audit** (an FPR or confusion structure) powers the analytic correction.
- It is the difference between “the model said so” and a **credible measurement**.
- One caveat: splitting and auditing fix *own-sample* and *error-rate* problems — not **domain shift** or systematic misclassification correlated with covariates.

Budget for a benchmark in *every* text project — even a small one.

Frontiers

Two Frontiers, Two Different Problems

Text is the outcome — Modarressi, Spiess & Venugopal (2025)

Detect whether treatment changes open-ended text, describe *how* it changes, validate those descriptions, and assess how much of the effect they capture.

Text is the treatment — Feldman, Venugopal, Spiess & Feder (2026)

Generate controlled variation in a latent textual feature, then estimate its effect without conditioning away the very treatment being studied.

Both combine representation learning with familiar econometric requirements.

Frontier 1: Text as Outcome

No Single Score Fits a Text Outcome

Modarressi, Spiess & Venugopal (2025)

An RCT collects open-ended responses Y_i after treatment D_i . We want to discover **how the responses changed**.

- A chosen score may miss the relevant change.
- Testing many words creates a multiple-testing problem.
- An LLM description may overfit the same responses used to test it.

We need to discover an interpretable effect without testing it on the same data.

Step 1: Detect Any Effect

Instead of predicting text from treatment, predict treatment from text:

$$\hat{D}_i = \hat{g}(Y_i).$$

If treatment does not change the distribution of responses, held-out text should not predict treatment assignment better than a trivial baseline.

The question

Does *anything* in the response distinguish treatment from control?

This detects an effect without committing to a narrow scalar outcome.

Sample Splitting Keeps Detection Honest

The predictor is flexible enough to overfit, so performance is evaluated on held-out responses.

1. Train the reverse predictor on the discovery sample.
2. Freeze the predictor before opening the test sample.
3. Evaluate held-out loss and use randomization / permutation inference.

This is Egami et al.'s principle applied to a different goal: first establish that the treatment changed text at all.

Step 2: Propose Testable Themes

Detection says that the groups differ, but not *how*.

On the training sample, an LLM:

- compares treatment and control responses;
- proposes simple themes that may explain the difference;
- produces a scoring rule for each proposed theme.

The LLM is used for **hypothesis generation**. Its themes are not yet findings.

Good Themes Must Satisfy Three Criteria

The target is a concise description of the *systematic difference* between groups, not a summary of every document.

1. **Distinct:** themes should not merely restate one another.
2. **Differentially complete:** together they should capture the dimensions along which treatment and control differ.
3. **Causally meaningful:** each theme should describe a plausible outcome, not an accidental textual artifact.

Step 3: Humans Validate on Held-Out Data

Proposed themes can be statistically unstable or substantively wrong.

- An expert scores held-out responses using the proposed themes.
- Treatment labels remain hidden while the expert scores the text.
- Only after scoring is frozen are treatment-control differences estimated.

The LLM proposes themes; held-out human labels decide whether they hold.

Step 4: Measure Completeness

A significant theme may explain only a small fraction of the textual effect. As in detection, we **again predict treatment assignment from text** — now asking how much of that signal a short description keeps.

Compare three held-out predictors of treatment assignment:

1. a trivial predictor using no text;
2. a predictor using only the validated theme scores;
3. a flexible predictor using the full text — the benchmark.

Completeness = the share of the way from (1) to (3) that the theme predictor (2) covers. That gap — how much better the *whole* text predicts treatment than no text — is the treatment's full footprint on the language; 100% means the themes match the full-text benchmark, 0% that they beat nothing.

Themes Recover Most of the Difference

Proof of concept: 200 arXiv econometrics abstracts, split by researcher interest

Read the **Completeness** column, not Accuracy: it rescales onto [0% = no text, 100% = full text] — the share of the detectable difference a description captures.

Predictor	Accuracy	Completeness
Trivial classification	71%	0%
Direct LLM classification	86%	100%
LLM themes, human labels	85%	93%
LLM themes, LLM labels	81%	67%
Supervised topic model	73%	13%

Human-validated themes recover **93%**; scored by the LLM alone, **67%**; a supervised topic model, **13%**. **Human validation is what makes the short description hold up.**

What the Outcome Pipeline Buys

detect → **describe** → **human-validate** → **measure completeness**

- The detection test does not depend on the LLM's interpretation.
- The interpretation is tested on data that did not generate it.
- Completeness states what the interpretable themes leave unexplained.

This is the class's two-threat framework assembled into one research design.

Frontier 2: Text as Treatment

What Aspect of Language Causes a Response?

Feldman, Venugopal, Spiess & Feder (2026)

Consider comments made by citizens at local-government meetings.

- Does using insults reduce agreement with the speaker?
- Does more civil language increase perceived legitimacy?
- Would the **same underlying message** have a different effect if this feature of its language changed?

The treatment is not the entire text — it is one **latent** feature (here, incivility) expressed through the language. This is the paper's own framing: a *latent textual treatment*.

Why This Is Hard to Turn into an Experiment

Ideally, compare two versions of a comment that differ only in the target feature:

$$\text{same message and context} \quad + \quad \begin{cases} \text{low use of insults,} \\ \text{high use of insults.} \end{cases}$$

But rewriting language usually changes several things at once: tone, length, stance, emotional intensity, or even the substantive message.

The first challenge is to create credible textual counterfactuals.

The Role of Labels: Which Feature to Steer

The SAE offers thousands of candidate concept directions.

Labels pick the one we mean — this step is **supervised** at its core.

- **Operationalize:** reduce the broad concept to a concrete feature — insults, one component of incivility.
- **Detect:** on texts labeled civil vs uncivil, keep the SAE feature whose activation best *separates* the two groups (mean difference, then an L_1 logistic probe ranked by persistence); confirm its meaning by hand.
- **Validate:** steer each candidate — keep those whose intensity rises *linearly* and whose text stays coherent.

The SAE narrows the search; the labels choose the feature; steering confirms it is a usable knob.

Generate Quasi-Counterfactual Texts

With the target feature chosen, a researcher uses an LLM to generate several versions of each base text, pushing that feature from low to high intensity.

These are **quasi-counterfactuals**: they aim to preserve the original context while changing the feature that will become the treatment.

To move *one* concept while leaving the rest, they isolate a single interpretable “concept direction” inside the LLM and dial it up or down (“steering”). The device is a sparse autoencoder — but treat it as a black-box **counterfactual generator**, not the econometric insight of the paper.

Did the Textual Manipulation Work?

The intended change cannot be taken for granted. For every generated text, check:

- **treatment intensity**: did the target feature actually move?
- **coherence**: is the text readable, relevant, and human-like?
- **support**: can the same base text produce both low- and high-intensity versions?

The realized treatment is the **measured intensity in the output**, not how strongly the researcher instructed the model to change it — a compliance gap, like dosage non-compliance: you set the dial, but you estimate the effect of the dose that actually landed.

The Remaining Problem: Rewriting Is Not Surgical

Even a successful manipulation may also change:

- tone, topic, length, specificity, or emotional intensity;
- the argument, stance, or information conveyed;
- artifacts introduced by the generator.

Randomly assigning generated texts to respondents solves selection into exposure. It does **not** guarantee that high- and low-intensity texts differ only in the target.

Why Not Control for the Rest of the Text?

Let T_i indicate high versus low realized intensity and X_i be an embedding of the generated text.

- X_i summarizes nuisance differences we would like to control for.
- But the target feature is itself part of the text, so a rich X_i also reveals treatment: $P(T_i = 1 \mid X_i) \in \{0, 1\}$.
- At a fixed full-text representation, there is no low- and high-treatment comparison. Overlap disappears.

Keep Context, Remove the Treatment Signal

The treatment (measured intensity) already exists; now we want **controls**, so we compare texts alike in everything *except* intensity.

Let X_i be a **dense vector summary of the whole text** — a sentence embedding, not the SAE feature (that one *measures* the treatment).

The catch: intensity is written into the text, so X_i also encodes the treatment. Control for *all* of X_i and the high-vs-low contrast vanishes — the positivity problem from the last slide.

Residualization: keep only the part of X_i that does *not* predict intensity ($X_i^\perp$), drop the part that does ($X_i^\parallel$). The controls then soak up topic and tone but no longer reveal intensity — FWL in reverse: partial the treatment *out of the controls*, leaving enough variation to still compare high vs low.

Estimate the Effect on Respondents

After constructing the residualized controls, estimate:

$$\tau(x) = E\left[Y(1) - Y(0) \mid \tilde{X} = x\right].$$

- whether respondents exposed to high-intensity text respond differently;
- while comparing texts with similar retained context;
- and allowing the effect to vary with that context.

The final estimation uses familiar causal-ML tools. The unusual part is constructing a control representation that does not condition away the textual treatment.

Residualization Works in Simulation

The paper generates steered texts from real corpora and assigns outcomes with known heterogeneous treatment effects.

- Full embeddings almost perfectly reveal treatment and perform poorly as controls.
- Residualized embeddings substantially reduce the gap between estimated and true effects.
- The best strategy depends on whether treatment information is concentrated in one direction or spread across dimensions.

The Two Frontier Pipelines Compared

	Text as outcome	Text as treatment
Scientific target	Distributional effect	Effect of a latent feature
Discovery tool	Reverse prediction + themes	LLM-generated text variants
Main threat	Adaptive description	Imprecise intervention
Design fix	Split + human validation	Residualize treatment from controls
Final object	Themes + completeness	ATE / CATE with overlap

Frontier methods do not suspend econometrics. They make measurement, splitting, overlap, and validation assumptions more visible.

Wrap-Up

Red Flags in a Referee Report

Signals that a text-as-data result needs a second look:

- “Validated at 95% accuracy” — with no account of how the error enters the *regression*. **Accuracy $\neq$ unbiased inference.**
- The measure was **discovered and tested on the same documents.**
- A generated regressor reported with plain OLS standard errors and **no bias correction** — right width, wrong center.
- Point estimates compared across corpora or languages as if g were portable — compare **signs**, not magnitudes.
- **No validation sample** budgeted at all.

Practical Checklist

1. State the role of text: outcome / regressor / treatment / control.
2. **Split**: discover and validate the measure on one set; estimate on another.
3. Build a **validation sample**; report accuracy, κ , confusion — or at least the FPR.
4. **Correct the bias**; report it is bias, not just SEs (ValidMLInference).
5. Archive: corpus, code, model version, prompt, and validation labels.

Course Takeaways

- Text is a tool to **measure** economic concepts at scale.
- Every measure — dictionary, classifier, topic, embedding, LLM label — is a **measurement with error**, and often a *discovered* one.
- **Validation, sample splitting**, and the **econometrics of generated variables** are what make text-as-data credible for inference.

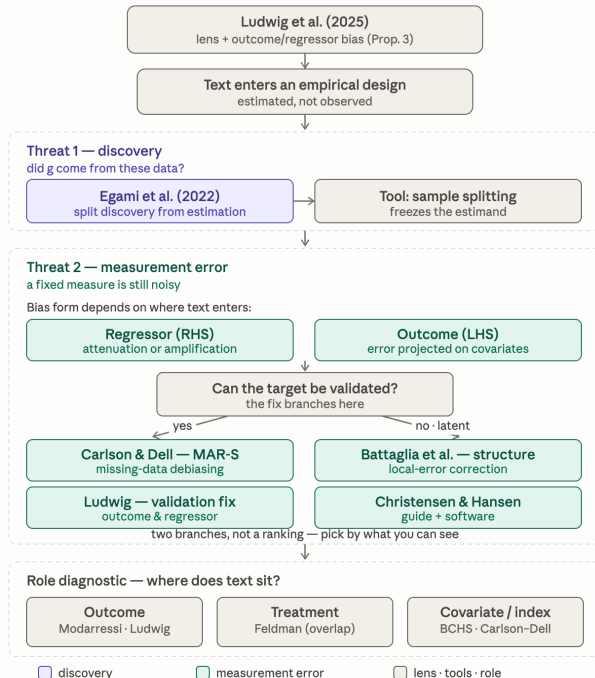
Mechanics: Problem Set 4 (classify → validate → regress → correct).

Core Readings

- Ludwig, Mullainathan & Rambachan (2025), *Large Language Models: An Applied Econometric Framework* — prediction, leakage, estimation, and validation-based debiasing.
- Egami, Fong, Grimmer, Roberts & Stewart (2022), *How to Make Causal Inferences Using Texts* — roles of text; discovery; sample splitting.
- Battaglia, Christensen, Hansen & Sacher (2025), *Inference for Regression with Variables Generated by AI/ML* — bias, valid SEs, corrections.

Correction Branches and Frontiers

- Carlson & Dell (2025; rev. 2026), *A Unifying Framework for Robust and Efficient Inference with Unstructured Data* — MAR-S and representative validation samples.
- Christensen & Hansen (2025) — applied guide and ValidMLInference software.
- Modarressi, Spiess & Venugopal (2025) — text as outcome.
- Feldman, Venugopal, Spiess & Feder (2026) — text as treatment.



☐ discovery

☐ measurement error

☐ lens · tools · role